

3rd Dec 2025
CSCI 6907

Automatic Phenotyping from Clinical Notes using MIMIC-IV

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What the Project is about?

- **Goal:** Develop an NLP-based system to automatically identify patients with specific diseases (heart failure and diabetes) using unstructured clinical notes from the MIMIC-IV dataset.
- **Approach:** Use BioClinical ModernBERT, a transformer-based model, to classify patients as positive or negative for both the diseases
- **Objective:** Evaluate the effectiveness of transformer-based NLP models in accurately phenotyping heart failure and diabetes patients using only unstructured clinical notes from MIMIC-IV.



Research Questions

RQ1 Can unstructured clinical notes accurately identify patients with heart failure and diabetes compared to ICD-based phenotyping?

RQ2 Which sections in the discharge summaries (e.g., history of present illness, past medical history, etc) help make the predictions?



Motivation

- Manual ICD coding is time-consuming and error-prone.
- Clinical notes contain richer, more detailed patient information than structured data.
- Automating phenotyping can improve research efficiency and clinical decision-making.



Dataset

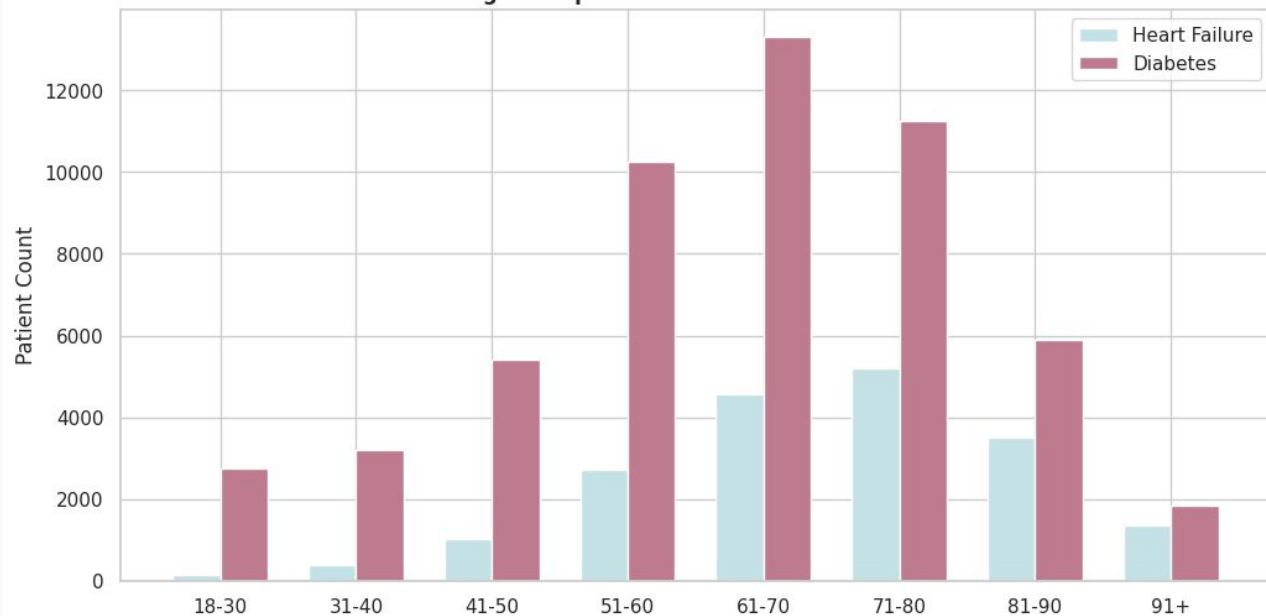
MIMIC-IV (Medical Information Mart for Intensive Care IV) — a large, publicly available, de-identified EHR dataset from MIT.

We are using data from:

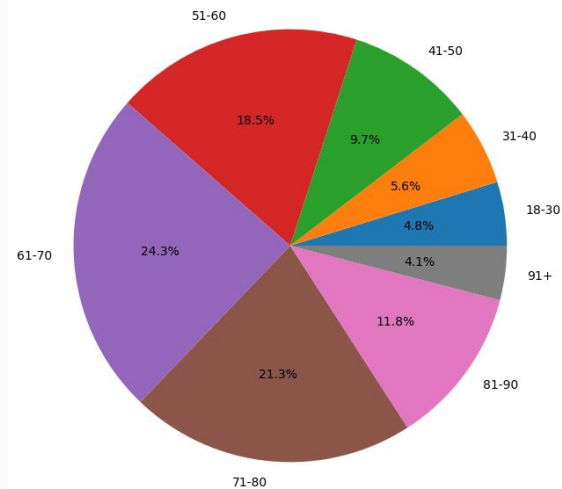
- | | | |
|---|--|--|
| 1 | Patients table | demographic details |
| 2 | Admissions table | hospitalization-level data |
| 3 | Diagnoses_ICD table | ICD-10 diagnosis codes used as phenotype labels |
| 4 | MIMIC-IV Note (Discharge Summaries) | unstructured clinical text containing detailed patient summaries |

Patient Demographics

Age Group Distribution: HF vs Diabetes

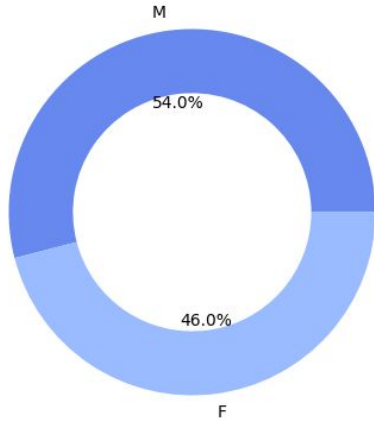


Age Group Distribution (Combined)

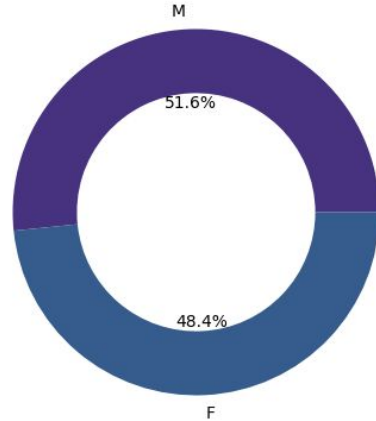


Patient Demographics

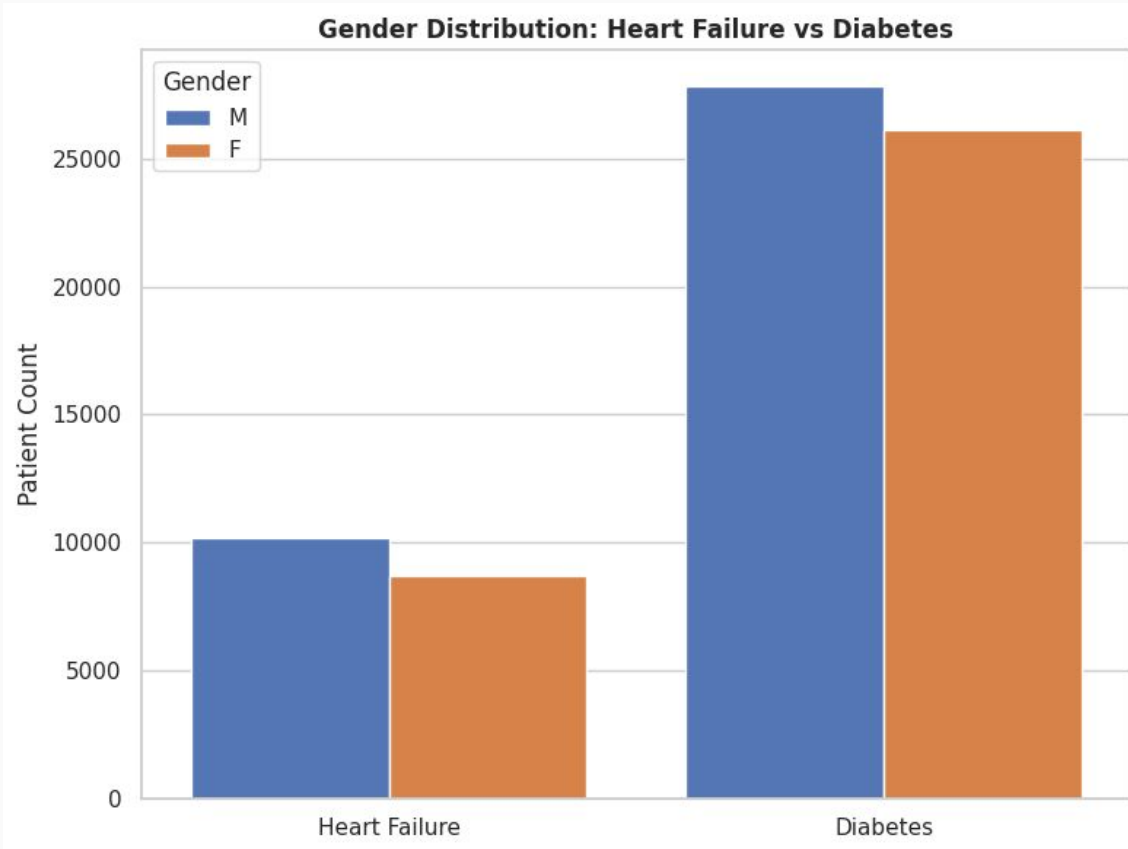
Gender Distribution - Heart Failure



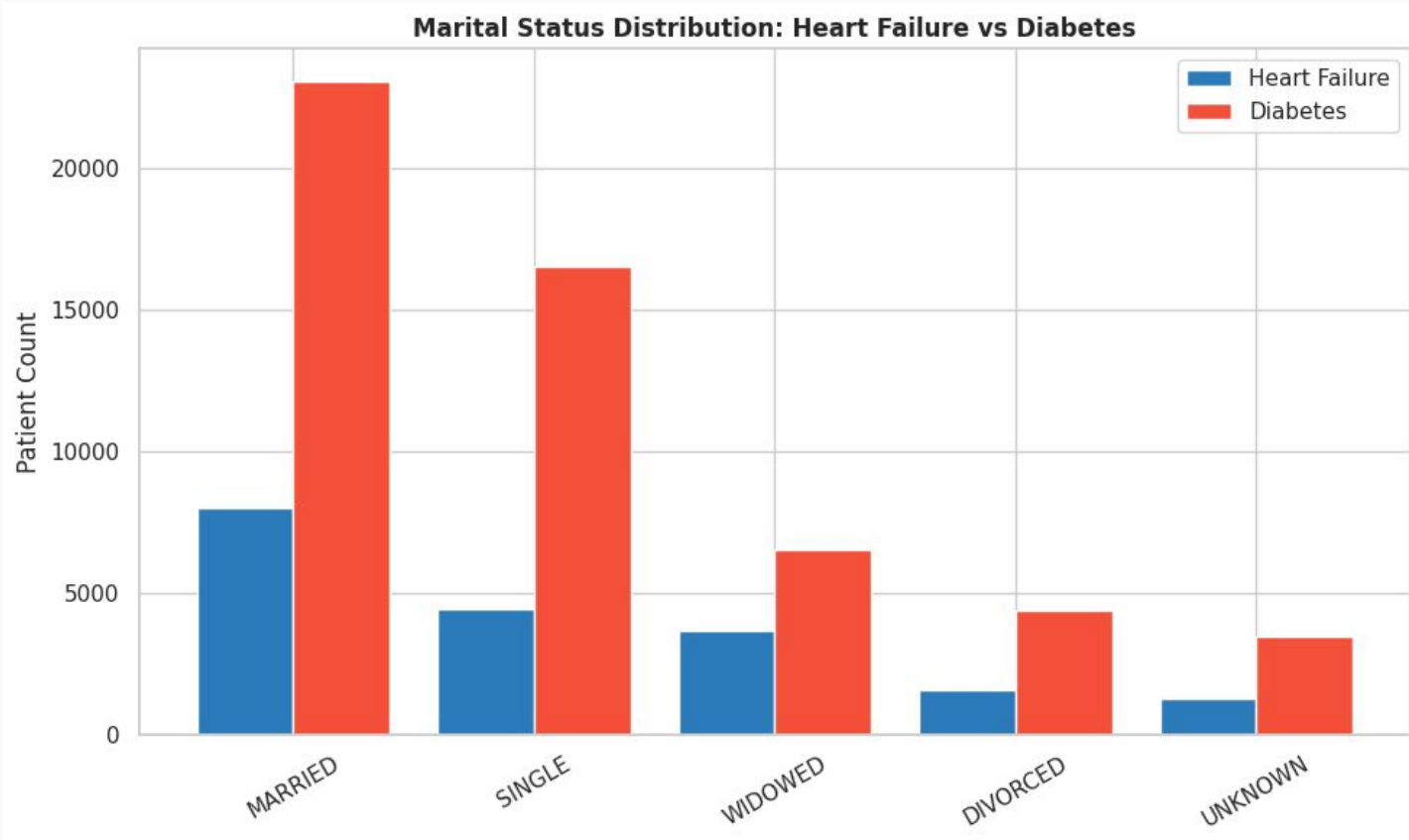
Gender Distribution - Diabetes



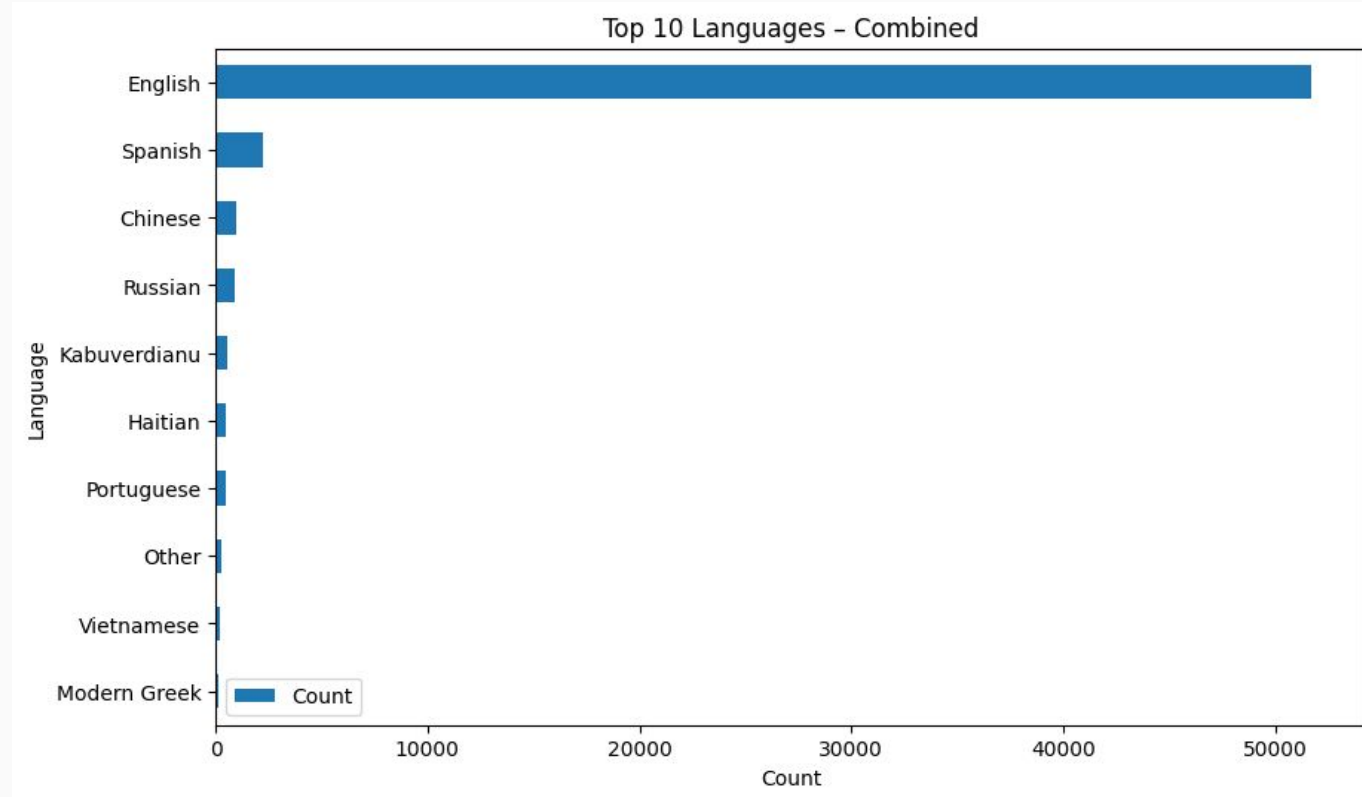
Patient Demographics



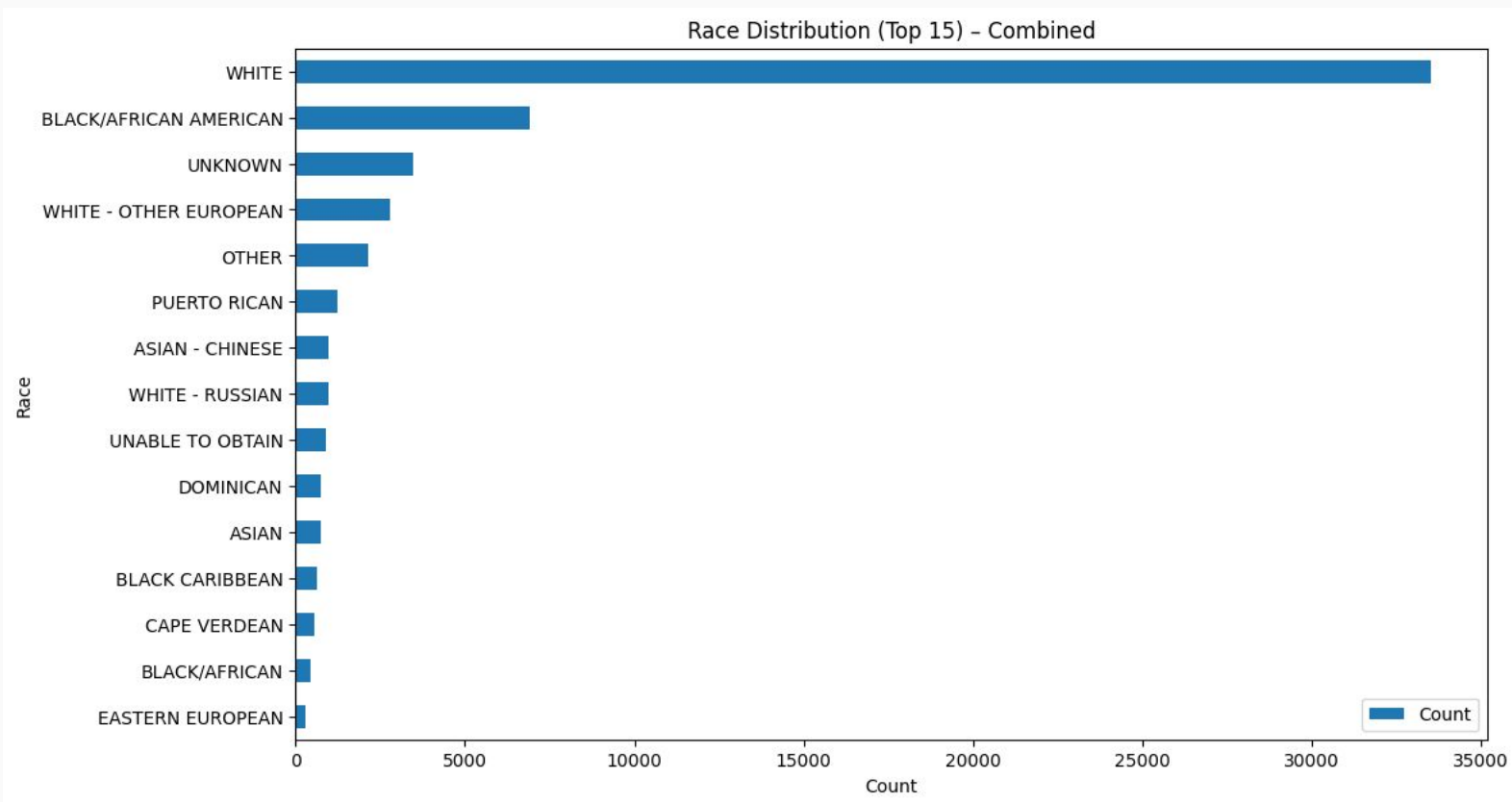
Patient Demographics



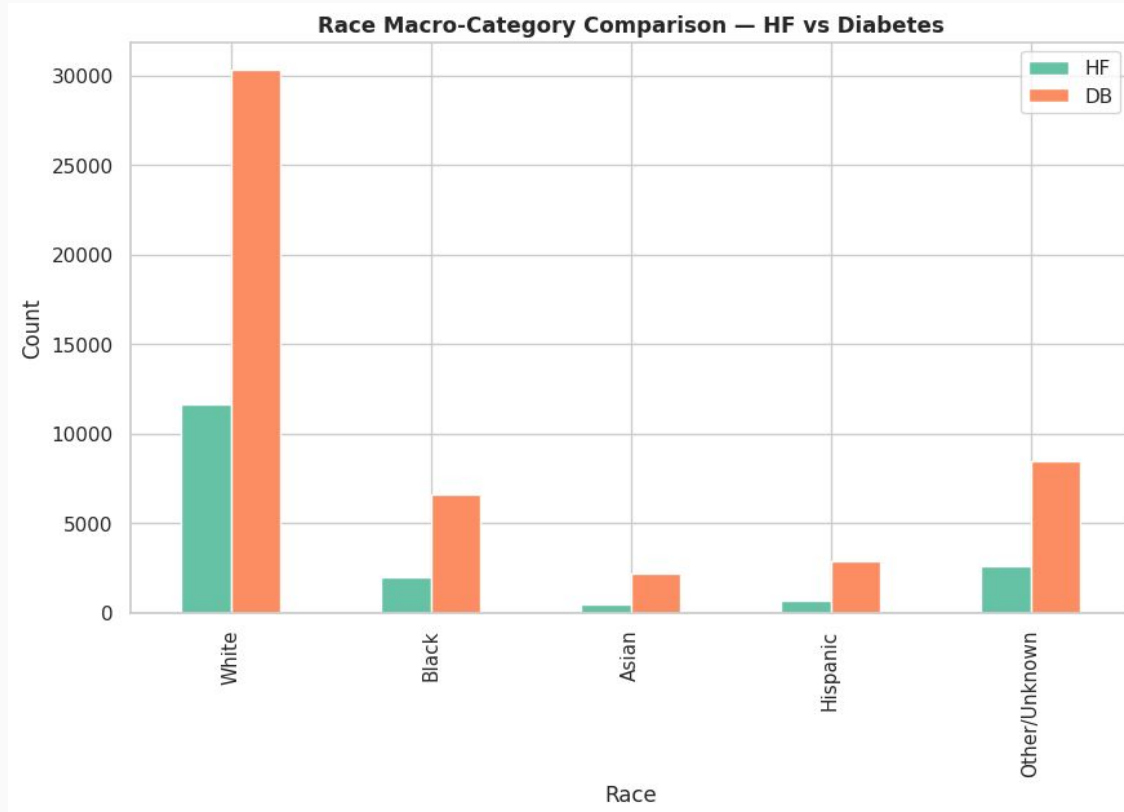
Patient Demographics



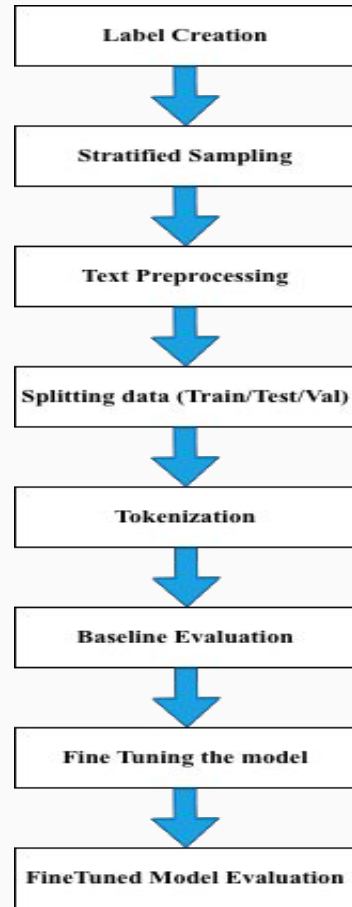
Patient Demographics



Patient Demographics



Workflow



Label Creation

- The targeted ICD-10 codes for the diseases were - “Heart Failure”: I50, “Diabetes Mellitus”: E8, E9, E10, E11, E13
- Merged the admissions, diagnoses_icd and discharge tables using subj_id and hadm_id
- Each corresponding Note had two labels for the two diseases
- They were labeled as 1 if there was a match or 0 if there wasn't a match.



Sampling

- Created a balanced set for phenotyping classification.
- Tried to ensure equal representation of HF only, Diabetes only, Both and None
- For each positive sub group -
 - Identified its size
 - Randomly sampled the same number of negatives from the “Neither” group
 - Removed sampled negatives so remaining ones can be used for the next class

```
Sampled dataset size: 128776
Label distribution (sum of 1s):
heartfailure_label    23451
diabetes_label        56976
dtype: int64
```



Text Preprocessing

- Prepared discharge summaries for downstream modeling.
- Clinical notes contain many unrelated sections
- Only the following sections were included -
 - History of Present Illness
 - Past Medical History
 - Hospital Course
 - Discharge Diagnosis
 - Discharge Medications
- Following the lowercasing of text, the following were removed -
 - Bracketed content
 - Extra Whitespace
 - Underscore runs
 - Excess Newlines



Splitting Data + Tokenization

- Created 3 splits - Train, Validation, Test
- The ratio was 70, 15, 15
- To convert cleaned clinical text into transformer-ready token embeddings.
- This prepares the model inputs for fine-tuning the BioClinical ModernBERT model
- The AutoTokenizer from the model was used
- Labels were converted into torch tensors. Each split (train, test, val) was saved as the following tensor -
 - Input_ids
 - Attention mask
 - Label



Training Setup

- The model was trained using an A100 GPU with 80 GB VRAM.
- The training parameters used are listed below -

```
training_args = TrainingArguments(  
    output_dir="../outputs",  
    eval_strategy="epoch",  
    save_strategy="epoch",  
    learning_rate=2e-5,  
    per_device_train_batch_size=112,  
    per_device_eval_batch_size=112,  
    num_train_epochs=5,  
    weight_decay=0.01,  
    logging_dir="../logs",  
    load_best_model_at_end=True,  
    metric_for_best_model="f1",  
    greater_is_better=True,  
    bf16=True,  
    optim="adamw_torch_fused",  
    gradient_checkpointing=True,  
    dataloader_num_workers=8,  
    dataloader_pin_memory=True,  
    max_grad_norm=1.0,  
    report_to="none",  
    logging_steps=100,  
)
```



Model Used - BioClinical MordernBERT

- The model that was fine-tuned is the BioClinical MordernBERT model
- It has 150M parameters.
- Trained on the largest clinical and biomedical corpus to date with over 53.5 billion tokens.
- It has also been trained on 20 different datasets from diverse institutions, domains and geographic regions.
- It was also trained on MIMIC 3 and MIMIC 4 notes.
- This model was picked as it supports long clinical narratives and and its token length is 4096



Data Validation

Due to the sensitive nature of MIMIC 4 data several data validation checks were performed.

They were

- Structural Inspection - Ensure saved objects only contain tensors.
- PHI/Raw Text Detection - Look for accidental strings.
- Tensor Summary Audit - Inspect the tensor statistics to make sure to confirm they represent token ID's and not text data.



Evaluation

	precision	recall	f1-score	support
Heart_Failure	1.00	0.00	0.00	3518
Diabetes	0.44	0.30	0.36	8547
micro avg	0.44	0.21	0.28	12065
macro avg	0.72	0.15	0.18	12065
weighted avg	0.61	0.21	0.25	12065
samples avg	0.13	0.11	0.12	12065

Baseline Model

Pre-trained BioClinical-MordernBERT



Evaluation

	precision	recall	f1-score	support
Heart_Failure	0.69	0.56	0.61	3518
Diabetes	0.68	0.88	0.77	8547
micro avg	0.68	0.78	0.73	12065
macro avg	0.69	0.72	0.69	12065
weighted avg	0.68	0.78	0.72	12065
samples avg	0.40	0.40	0.39	12065

Fine-tuned BioClinical ModernBERT

Evaluation

Modern Pubmed-BERT - 0.1B parameters

pre-trained

	precision	recall	f1-score	support
Heart_Failure	0.19	0.72	0.30	3518
Diabetes	0.45	0.93	0.61	8547
micro avg	0.34	0.87	0.48	12065
macro avg	0.32	0.82	0.45	12065
weighted avg	0.37	0.87	0.52	12065
samples avg	0.33	0.44	0.36	12065

fine-tuned

	precision	recall	f1-score	support
Heart_Failure	0.70	0.57	0.63	3518
Diabetes	0.73	0.78	0.76	8547
micro avg	0.73	0.72	0.72	12065
macro avg	0.72	0.67	0.69	12065
weighted avg	0.72	0.72	0.72	12065
samples avg	0.36	0.36	0.35	12065

Evaluation

Metrics from similar papers

Paradigm	Model	Setting	Entity	Exact			Relaxed		
				Precision	Recall	F1	Precision	Recall	F1
Pre-train + Fine-tune	BioClinicalBERT	Supervised	Rare disease	0.689	0.720	0.704	0.772	0.739	0.755
			Disease	0.494	0.488	0.491	0.532	0.538	0.535
			Sign	0.561	0.516	0.538	0.676	0.735	0.704
			Symptom	0.667	0.630	0.648	0.704	0.745	0.724
			Overall	0.600	0.583	0.591	0.681	0.698	0.689

Model	Classification
<i>Non-medical models</i>	
BAAI Bge Base En V1.5	0.69 ± 0.22
AllenAI Scibert Scivocab Uncased	0.60 ± 0.20
Google BERT Base Uncased	0.58 ± 0.20
Intfloat E5 Base	0.68 ± 0.22
Nomic AI Nomic Embed Text V1	0.69 ± 0.22
Nomic AI Nomic Embed Text V1 Unsupervised	0.69 ± 0.22
Sentence-Transformers All MiniLM L6 V2	0.68 ± 0.21
Sentence-Transformers All MPNet Base V2	0.70 ± 0.22
Thenlper GTE Base	0.70 ± 0.22
<i>Medical models</i>	
Abhinand MedEmbed Base	0.69 ± 0.22
BioNLP Bluebert PubMed Mimic Uncased L-12 H-768 A-12	0.62 ± 0.21
EmilyAlsentzer Bio ClinicalBERT	0.59 ± 0.21
Kamalkraj BioSimCSE BioLinkBERT Base	0.64 ± 0.22
Malteos SciNCL	0.69 ± 0.22
MedicalAI ClinicalBERT	0.60 ± 0.21
Microsoft BiomedNLP BiomedBERT Base Uncased Abstract Fulltext	0.61 ± 0.20
MedTE C115 Step 8000	0.72 ± 0.23

Limitations

- Fine-tuned on data from a single source so this limits generalizability to reports from other hospitals
- No integrated into EHR workflows, so performance in a real world setting remains unknown
- Due to limited hardware and compute, we weren't able to run many training experiments.





The End